

4.1 Maclaurin & Taylor Polynomials Worksheet

1. Given a function $f(x)$ for which

$$f(2) = 3, \quad f'(2) = 5, \quad f''(2) = 8, \quad f'''(2) = -18, \quad \& \quad f^{(4)}(2) = 240,$$

write out the degree 4 Taylor polynomial for $f(x)$ centered at $x = 2$.

$$\text{We want to find: } p_4 = \frac{f(2)}{0!} + \frac{f'(2)}{1!}(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3 + \frac{f^{(4)}(2)}{4!}(x-2)^4$$

Using the information we are given in the problem we see that:

$$p_4 = 3 + 5(x-2) + 4(x-2)^2 - 3(x-2)^3 + 10(x-2)^4$$

2. Find the 3rd Taylor polynomial (that is, find p_3) for $f(x) = \sqrt{x}$ about $x = 4$.

$$\text{We want to find: } p_3 = \frac{f(4)}{0!} + \frac{f'(4)}{1!}(x-4) + \frac{f''(4)}{2!}(x-4)^2 + \frac{f'''(4)}{3!}(x-4)^3$$

$$f(x) = \sqrt{x} \quad \rightarrow \quad f(4) = 2$$

$$f'(x) = \frac{1}{2\sqrt{x}} \quad \rightarrow \quad f'(4) = \frac{1}{4}$$

We note that

$$f''(x) = \frac{-1}{4x^{3/2}} \quad \rightarrow \quad f''(4) = \frac{-1}{32}$$

$$f'''(x) = \frac{3}{8x^{5/2}} \quad \rightarrow \quad f'''(4) = \frac{3}{256}$$

Therefore, the 3rd Taylor polynomial is:

$$p_3 = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2 + \frac{1}{512}(x-4)^3$$

3. Consider the function $f(x) = \tan^{-1}(x)$.

a) Find the 3rd Maclaurin polynomial for $f(x) = \tan^{-1}(x)$.

$$\text{We want to find: } p_3 = \frac{f(0)}{0!} + \frac{f'(0)}{1!}(x-0) + \frac{f''(0)}{2!}(x-0)^2 + \frac{f'''(0)}{3!}(x-0)^3$$

$$f(x) = \tan^{-1}(x) \rightarrow f(0) = 0$$

$$f'(x) = \frac{1}{x^2 + 1} \rightarrow f'(0) = 1$$

We note that $f''(x) = \frac{-2x}{(x^2 + 1)^2} \rightarrow f''(0) = 0$

$$f'''(x) = \frac{-2(x^2 + 1)^2 + 8x^2(x^2 + 1)}{(x^2 + 1)^4} \rightarrow f'''(0) = -2$$

Therefore, the 3rd Maclaurin polynomial is:

$$p_3 = 0 + x + 0 - \frac{1}{3}x^3 = x - \frac{1}{3}x^3$$

b) Use the polynomial you found in part (a) to provide an estimate of the value $\tan^{-1}(0.25)$.

We now that in part (a) the value of $p_3(0.25) \approx \tan^{-1}(0.25)$ since $x = 0.25$ is close to $x = 0$.

$$\text{Therefore } \tan^{-1}(0.25) \approx (0.25) - \frac{1}{3}(0.25)^3 = \frac{1}{4} - \frac{1}{3}\left(\frac{1}{64}\right) = \frac{47}{192} \approx 0.2448$$

Note that the actual value of $\tan^{-1}(0.25)$ is roughly 0.2250

4. What is the coefficient of x^2 in the Maclaurin polynomial (which has degree 2 or higher) for

$$f(x) = \frac{x-2}{6x^2 + x - 1} ?$$

We know that the coefficient of the x^2 term in the polynomial will be $\frac{f''(0)}{2!}$. We note that

$$f'(x) = \frac{6x^2 + x - 1 - (x-2)(12x+1)}{(6x^2 + x - 1)^2} = \frac{6x^2 + x - 1 - (12x^2 - 23x - 2)}{(6x^2 + x - 1)^2} = \frac{-6x^2 + 24x + 1}{(6x^2 + x - 1)^2} \text{ and that}$$

$$f''(x) = \frac{(6x^2 + x - 1)^2(-12x + 24) - 2(6x^2 + x - 1)(12x + 1)(-6x^2 + 24x + 1)}{(6x^2 + x - 1)^4}.$$

$$\text{Therefore we have that } f''(0) = \frac{(-1)^2(24) - 2(-1)(1)(1)}{1} = 26.$$

$$\text{Thus the coefficient of the } x^2 \text{ term in the polynomial is } \frac{f''(0)}{2!} = \frac{26}{2} = 13.$$

5. Use a Taylor polynomial of degree 2 to obtain a good estimate for $\ln(1.2)$. You don't need to simplify your answer.

We can approximate $\ln(1.2)$ by creating a Taylor polynomial for the function $f(x) = \ln(x)$ about $x = 1$. Then since $x = 1.2$ is close to $x = 1$ we will know that $p_2(1.2) \approx \ln(1.2)$

$$\text{We want to find: } p_2 = \frac{f(1)}{0!} + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2$$

$$f(x) = \ln(x) \rightarrow f(1) = 0$$

$$\text{We note that } f'(x) = \frac{1}{x} \rightarrow f'(1) = 1$$

$$f''(x) = -\frac{1}{x^2} \rightarrow f''(1) = -1$$

Therefore, the 2nd Taylor polynomial is:

$$p_2 = 0 + (x-1) - \frac{1}{2}(x-1)^2$$

Therefore

$$\ln(1.2) \approx p_2(1.2) = (1.2-1) - \frac{1}{2}(1.2-1)^2 = 0.2 - \frac{1}{2}(0.2)^2 = 0.2 - \frac{1}{2}(0.04) = 0.2 - 0.02 = 0.18$$

Note that the actual value of $\ln(1.2)$ is roughly 0.1823.

6. Use a Taylor polynomial of degree 2 to approximate the value of $\sqrt{103}$.

We can approximate $\sqrt{103}$ by creating a Taylor polynomial for the function $f(x) = \sqrt{x}$ about $x = 100$. Then since $x = 103$ is close to $x = 100$ we will know that $p_2(103) \approx \sqrt{103}$

$$p_2 = \frac{f(100)}{0!} + \frac{f'(100)}{1!}(x-100) + \frac{f''(100)}{2!}(x-100)^2$$

$$f(x) = \sqrt{x} \rightarrow f(100) = 10$$

$$\text{We note that } f'(x) = \frac{1}{2\sqrt{x}} \rightarrow f'(100) = \frac{1}{20}$$

$$f''(x) = \frac{-1}{4x^{3/2}} \rightarrow f''(100) = \frac{-1}{4000}$$

Therefore, the 2nd Taylor polynomial is:

$$p_2 = 10 + \frac{1}{20}(x-100) - \frac{1}{8000}(x-100)^2$$

Therefore

$$\begin{aligned}\sqrt{103} \approx p_2(103) &= 10 + \frac{1}{20}(103-100) - \frac{1}{8000}(103-100)^2 \\ &= 10 + \frac{3}{20} - \frac{9}{8000} \\ &= \frac{81191}{8000} \approx 10.148875\end{aligned}$$

Note that the actual value of $\sqrt{103}$ is roughly 10.14889157.